Radian Measure and Arc Length

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Converting between degrees and radians

Convert each angle to radians, in exact form:

a 45° **b** 120° **c** 270° **d** 330°

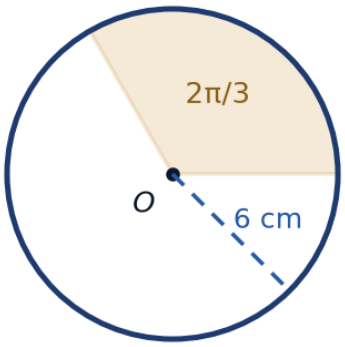
2 [Foundational] • Target: Converting between degrees and radians

Convert each angle to degrees. Round to one decimal place where needed:

a $\frac{\pi}{6}$ **b** $\frac{5\pi}{4}$ **c** $\frac{7\pi}{3}$ **d** 3.5 rad

3 [Foundational] • Target: Arc length and sector area

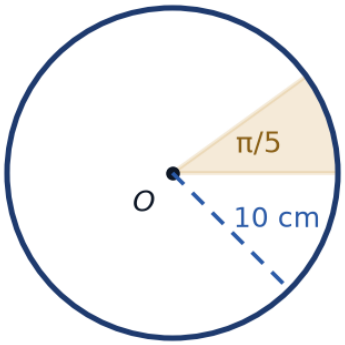
A circle has radius 6 cm. Find the exact length of the arc subtended by a central angle of $\frac{2\pi}{3}$.



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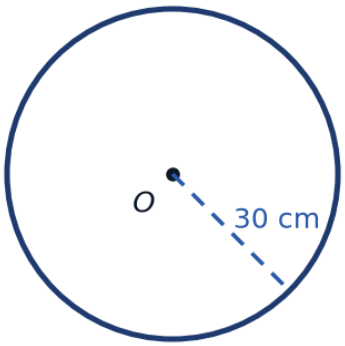
4 [Foundational] • Target: Arc length and sector area

A sector of a circle of radius 10 cm has central angle $\frac{\pi}{5}$. Find its exact area.



5 [Foundational] • Target: Arc length and sector area

A wheel of radius 30 cm rolls along the ground without slipping.



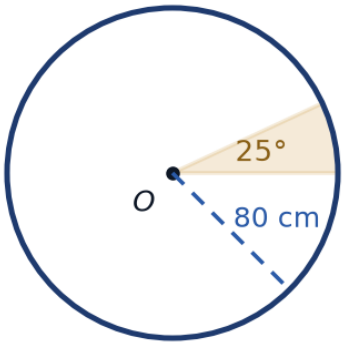
a Through what angle, in radians, does the wheel turn when it travels 1.5 m?

b Express this angle to the nearest tenth of a degree.

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6 [Foundational] • Target: Arc length and sector area

A pendulum of length 80 cm swings through an angle of $25°$. Find the length of the arc traced by its tip, to the nearest tenth of a centimetre.



7 [Foundational] • Target: Angular and linear speed

A carousel horse is 4 m from the center and completes one full rotation every 20 seconds.

a Find the angular speed of the horse, in radians per second.

b Find the linear speed of the horse, in m/s.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Angular and linear speed

A car's wheel has radius 35 cm and spins at 400 revolutions per minute. Find the car's speed in km/h, to one decimal place.

9 [Intermediate] • Target: Angular and linear speed

Two gears mesh together: the smaller gear has radius 5 cm and spins at 300 rpm. If the larger gear has radius 15 cm, find its rotational speed in rpm (the linear speed at the point of contact is the same for both gears).

10 [Intermediate] • Target: Coterminal angles and reference angles

Find a positive coterminal angle (between 0 and 2π) for each:

a $\frac{9\pi}{4}$ **b** $-\frac{\pi}{3}$ **c** $\frac{13\pi}{6}$

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11 [Intermediate] • Target: Coterminal angles and reference angles

Find the reference angle for each angle:

a $\frac{5\pi}{6}$ **b** $\frac{7\pi}{4}$ **c** $\frac{4\pi}{3}$

12 [Intermediate] • Target: The radian definition itself

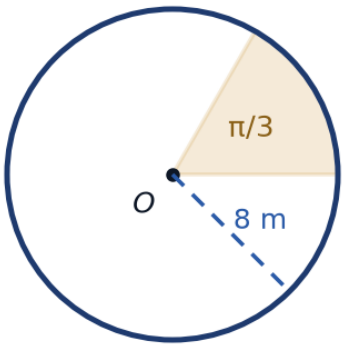
State the definition of one radian in terms of arc length and radius, and explain why radians are considered a 'natural' unit for angles.

13 [Intermediate] • Target: The radian definition itself

Explain why the full circle corresponds to 2π radians, using the circumference formula $C = 2\pi r$.

14 [Intermediate] • Target: Combined arc length and sector area problems

A sprinkler sprays water over a sector of radius 8 m and central angle $\frac{\pi}{3}$. Find both the area watered and the length of the outer arc of the watered region.

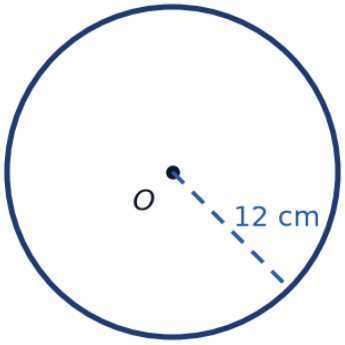


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Combined arc length and sector area problems • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A slice of pizza (a sector) has radius 12 cm and arc length 10 cm along its crust. Find the central angle in radians and the area of the slice.



16 [Challenging] • Target: Common angles in both units (reference table) • Cross-Lesson: 7.02 Logarithms and Their Properties

Complete a conversion table for the common angles $0°, 30°, 45°, 60°, 90°, 180°, 360°$ in radians.

17 [Challenging] • Target: Satellite and orbital speed applications • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its angular speed in rad/h and its linear speed in km/h, to the nearest whole number.

18 [Challenging] • Target: Satellite and orbital speed applications • Cross-Lesson: 7.04 Applications: Growth and Decay

Earth rotates once every 24 hours. Find the linear speed (due to rotation alone) of a point on the equator, using Earth's radius ≈ 6371 km, to the nearest km/h.

19 [Challenging] • Target: Negative angles and full-turn arcs • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the exact arc length swept by a radius-4 circle for a rotation of $-\frac{3\pi}{2}$ radians (clockwise), and interpret the sign.

20 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 8.02 Graphing Rational Functions

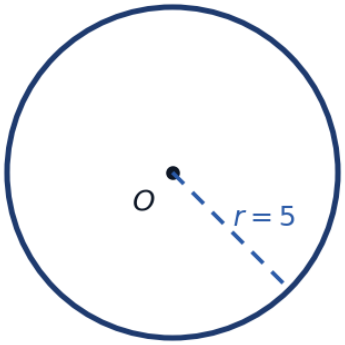
In a circle of radius 10 cm, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting between degrees and radians

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.



a. r = 9, $\theta = \frac{2\pi}{3}$

b. r = 6, $\theta = \pi$

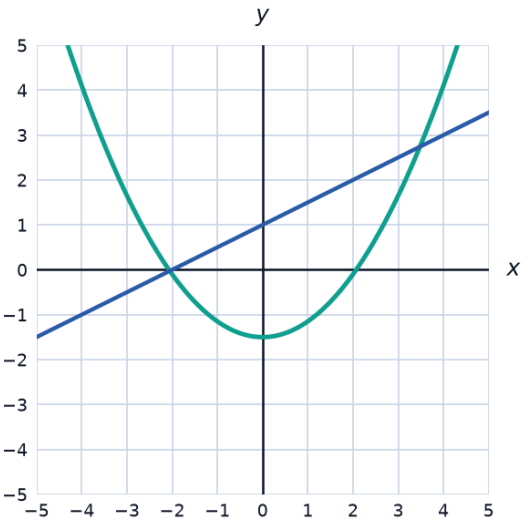
c. r = 3, $\theta = 2\pi$

d. r = 18, $\theta = \frac{\pi}{3}$

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22 [Foundational] • Target: Arc length and sector area

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is 6 cm and the radius is 3 cm, find the angle θ in radians.



a. $\frac{1}{2}$ radians

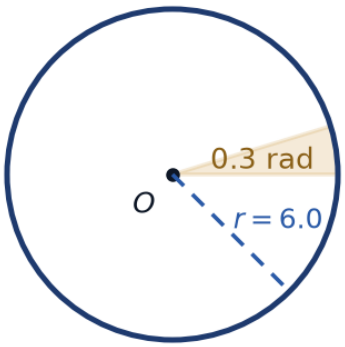
b. 2 radians

c. $\frac{2}{3}$ radians

d. $\frac{3}{2}$ radians

23 [Foundational] • Target: Angular and linear speed

A sector has radius 6 cm and angle 0.3 rad. Find the arc length.



a. 9/5

b. 63/10

c. 14/5

d. 27/5

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24 [Foundational] • Target: Coterminal angles and reference angles

Convert $\frac{\pi}{3}$ radians to degrees.

a. 300°

b. 60°

c. 90°

d. 30°

25 [Foundational] • Target: The radian definition itself

Consider a particle whose position in the complex plane at time t (in seconds) is given by $z(t) = 3e^{i(\frac{\pi}{6}t + \frac{\pi}{3})}$. (a) Find the initial position of the particle at $t = 0$ in polar coordinates (r, θ), where $r \geq 0$ and $0 \leq \theta < 2\pi$. (b) Determine the angular velocity of the particle in radians per second. (c) If the particle starts at a point A and moves along a circular path, find the arc length traveled by the particle from $t = 0$ to $t = 4$ seconds.

a. (a) $(3, \frac{\pi}{3})$, (b) $\frac{\pi}{6}$ rad/s, (c) 2π

b. (a) $(3, \frac{\pi}{6})$, (b) $\frac{\pi}{3}$ rad/s, (c) $\frac{4\pi}{3}$

c. (a) $(3, \frac{2\pi}{3})$, (b) $\frac{\pi}{6}$ rad/s, (c) $\frac{2\pi}{3}$

d. (a) $(3, \frac{\pi}{3})$, (b) $\frac{\pi}{3}$ rad/s, (c) $\frac{8\pi}{3}$

26 [Foundational] • Target: Combined arc length and sector area problems

Find the arc length of a sector with radius 5 and angle 2 radians.

a. 25

b. π

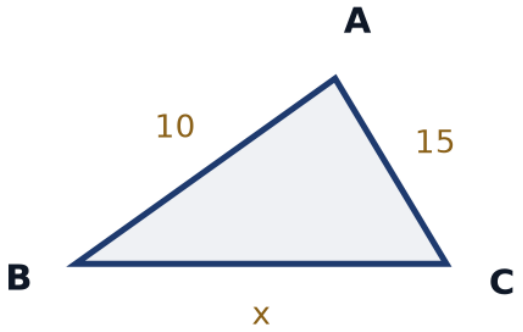
c. 5

d. 10

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27 [Foundational] • Target: Common angles in both units (reference table)

In triangle ABC, angle ABC = θ degrees, and angle ACB = 2θ degrees. The lengths of sides AB and AC are 10 cm and 15 cm respectively. The length of BC is denoted by x cm. Given that the area of triangle ABC is 60 cm^2, find the value of x, to 3 significant figures.



a. 13.7

b. 11.4

c. 12.8

d. 14.9

28 [Foundational] • Target: Satellite and orbital speed applications

Let the complex number z_1 be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, z_2, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of θ_2 in radians, given that $0 \le \theta_2 < 2\pi$.

a. $\frac{\pi}{2}$

b. $\frac{2\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{6}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Negative angles and full-turn arcs

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. 1

d. 5/3

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30 [Intermediate] • Target: Converting between degrees and radians

Consider an arithmetic sequence where the 5th term is 19 and the 11th term is 43. If the first term is denoted by a_1 and the common difference by d, find the value of $a_1 + d$.

a. 6

b. 7

c. 8

d. 9

31 [Intermediate] • Target: Arc length and sector area

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the 4th term of this sequence, a_4?

a. 11

b. 14

c. 17

d. 20

32 [Intermediate] • Target: Angular and linear speed

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fourth term of this sequence, a_4?

a. 14

b. 11

c. 17

d. 8

33 [Intermediate] • Target: Coterminal angles and reference angles

Find the angle between $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle 4, 2 \rangle$ (to one decimal place).

a. 0.0°

b. 20.0°

c. 180.0°

d. 10.0°

34 [Intermediate] • Target: The radian definition itself

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 - x + 4$.

a. $4x^2 - 1$

b. $8x + 4$

c. $4x - 1$

d. $8x - 1$

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35 [Intermediate] • Target: Combined arc length and sector area problems

Find the 4th term of the arithmetic sequence with $a_1 = 6$ and common difference $d = -5$.

a. -4

b. -19

c. -14

d. -9

36 [Intermediate] • Target: Common angles in both units (reference table)

Find a positive coterminal angle with 400° in $[0^\circ, 360^\circ)$.

a. 30

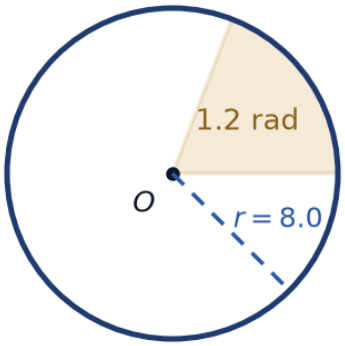
b. 40

c. 31

d. 400

37 [Intermediate] • Target: Satellite and orbital speed applications

A sector OAB has radius 8 cm and angle $\angle AOB = 1.2$ rad. Find the area of the minor segment cut off by chord AB, leaving your answer in exact form.



a. $32(1.2 - \sin(1.2))$

b. $64(1.2 - \sin(1.2))$

c. $33(1.2 - \sin(1.2))$

d. $32(1.2 + \sin(1.2))$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Negative angles and full-turn arcs • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the reference angle of 120°.

a. 120

b. 60

c. 31

d. 30

39 [Challenging] • Target: Converting between degrees and radians • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the angle between $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle 0, 1 \rangle$.

a. 135°

b. 60°

c. 90°

d. 120°

40 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 8.02 Graphing Rational Functions

For the ellipse $\frac{X^2}{81} + \frac{Y^2}{9} = 1$, find the length of the major axis and the foci.

a. Major axis 18, foci $(\pm\sqrt{72}, 0)$

b. Major axis 9, foci $(\pm\sqrt{72}, 0)$

c. Major axis 18, foci $(0, \pm\sqrt{72})$

d. Major axis 6, foci $(\pm\sqrt{72}, 0)$

41 [Challenging] • Target: Angular and linear speed • Cross-Lesson: 8.03 Introduction to Limits

Find the common ratio of the geometric sequence $2, 6, 18, 54, \ldots$.

a. 3

b. 2

c. -3

d. 4

42 [Challenging] • Target: Coterminal angles and reference angles • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Find the area of the polar region bounded by $r = 2$ for $0 \le \theta \le \frac{\pi}{6}$.

a. $\frac{2\pi}{3}$

b. $\frac{\pi}{3}$

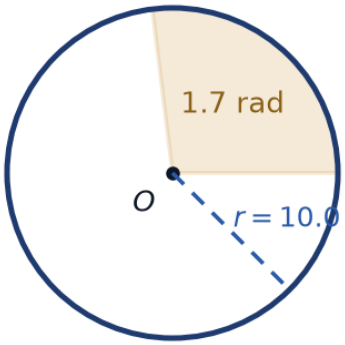
c. $\frac{\pi}{6}$

d. $\frac{4\pi}{3}$

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43 [Challenging] • Target: The radian definition itself • Cross-Lesson: 9.02 Geometric Sequences and Series

A sector has radius 10 cm and angle 1.7 rad. Find its area.



a. 170

b. 95

c. 85

d. 17

44 [Challenging] • Target: Combined arc length and sector area problems • Cross-Lesson: 9.03 The Binomial Theorem

A geometric sequence has common ratio $r = 3$ and $a_4 = 54$. Find a_1.

a. 5

b. 54

c. 2

d. 18

45 [Challenging] • Target: Common angles in both units (reference table) • Cross-Lesson: 1.01 Functions and Function Notation

For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, what is the length of the major axis?

a. 6

b. 10

c. 5

d. 8

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